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DCCN Lab Assignment-1

(1) Intermediate devices are networking components that help data move from a source to a destination across networks. They manage, control, filter and route data so that communication is efficient, reliable and secure. The major intermediate devices are:-

1) MODEM

A modem (MOdulator - DEModulator) allows digital devices to communicate over analog transmission media such as telephone lines or cable systems. It converts digital data into analog signals for transmission and reconverts incoming analog signals into digital form. Modems are commonly used to connect a home ~~*~~ or office network to an Internet Service Provider (ISP).

2) HUB

A hub is a simple networking device that connects multiple computers in a network. It receives data from one device and broadcasts it to all other connected devices.

3) REPEATER

A repeater is used to extend the range of a network. It receives weak or distorted signals, regenerates them, and sends them forward at full strength. This prevents signal degradation over long distances. Repeaters operate only at the physical layer.

4) SWITCH

A switch connects multiple devices within a LAN and forwards data only to the intended devices. It identifies devices using their MAC addresses and sends frames only to the correct port. This greatly improves network efficiency and reduces unnecessary traffic compared to hubs.

5) BRIDGE

A bridge connects two or more network segments and manages traffic between them. It filters data using MAC addresses and allow only necessary data to pass between segmentation. This reduces congestion and improves overall network performance.

6) ROUTER

A router connects different networks and directs data packets between them using IP addresses. It determines the best path for data to travel using routing tables. Routers are responsible for connecting local networks to the internet.

7) FIREWALL

A firewall is a network device for controlling network security and access rules. Firewalls are typically configured to reject access requests from unrecognised sources while allowing actions from recognised ones

8) GATEWAY

A gateway is a device or software that connects different networks using different protocols. It acts as a translator, converting data formats and communication rules so systems can understand each other.

(2)

1. arp :- displays or modifies ARP table (IP ⟷ MAC) mapping
2. ping :- checks connectivity b/w two network devices
3. ipconfig / ifconfig :- displays IP configuration of the system
4. netstat :- show n/w connections, routing table & statistics
5. traceroute / tracepath :- displays the path packets take to reach a destination.
6. route :- shows or modifies the system routing table
7. ftp :- transfers files b/w systems using File Transfer Protocol (insecure)
8. sftp :- securely transfers files using SSH
9. scp :- securely copies files using SSH
10. ssh :- establishes a secure remote login session to another machine
11. telnet :- connects to a remote host using an insecure text-based protocol
12. rlogin :- logs into a remote Unix machine
13. host :- retrieves DNS info about a domain or an IP.
14. whois :- fetches registration details of domain name
15. nslookup :- queries DNS servers to rescue domain names
16. xhost :- controls access to the X window display.
17. xclock :- displays a graphical clock.